

## BASIC RESEARCH

# The modeling study of the effect of morphological behaviors of extracellular matrix fibers on the dynamic interaction between tumor cells and antitumor immune response

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## Abstract

Low response rate limits the effective application of immunotherapy, in which the interactions between tumor cells and immune cells play a significant role. The strength of regulation could be mediated by extracellular matrix (ECM) fibers, which is still insufficiently investigated. In the study, the cellular potts model was utilized to explore the role of morphological properties of ECM in tumor-immune interactions. It was observed that high-density random ECM fibers delayed the interaction between tumor cells and T cells. Moreover, the tumor-immune interactions were ECM morphology-specific. Radial ECM fibers exhibited weaker inhibitory role in the process of contact between tumor cells and T cells. This study provided the useful mechanism of tumor-immune interactions from the viewpoint of morphological effect of ECM fibers, facilitating improving the efficiency of immunotherapy.

## KEYWORDS

ECM density, ECM morphology, immunotherapy, tumor-immune interaction

## 1 | INTRODUCTION

Regulation of tumor development is a complex task corresponding to variable tumor types. The immune system is involved with the recognition, killing, and elimination of tumor cells.<sup>1</sup> The self-protection mechanism of tumor cells could suppress the capability of cytotoxic T cells via PD-L1, CTLA-4, and so forth, leading to incomplete elimination of tumor cells. Up-regulation of PD-L1 levels of Tregs suppressed the proliferation of T cells in glioblastoma.<sup>2</sup> The inhibitory function of combination of PD-1/PD-L1 to T cells could be significantly relieved by anti-PD-1 immune checkpoint inhibitors, such as nivolumab.<sup>3</sup> The efficiency of antitumor immune response was variable. Further understanding the

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mechanism of interactions between the tumor and the immune system will further facilitate inhibition of tumor development and treatment of cancers.

Heterogeneity of extracellular matrix (ECM) has been hypothesized to play a vital role in regulating behaviors of tumor cells and T cells. The morphology of ECM includes patterns of distribution and density. Different arrangement of ECM fibers could be observed including random, aligned, and radial patterns.<sup>4</sup> ECM fibers were commonly randomly distributed. Cells at the borders of the tumor lump arranged random fibers into radial ones.<sup>5</sup> Crosslinking enzymes secreted by breast malignancy regulated changes in fiber patterns from random to aligned.<sup>6</sup> Numerous studies demonstrated that the morphological remodeling of ECM in tumor microenvironment was a signature of tumor progression. Cylindrical symmetry found in breast carcinoma tissue could be observed compared with helicoidal supramolecular assembly of collagen fibers found in perilesional tissue.<sup>7,8</sup> Straightened and aligned collagen fibers could determine a poor survival rate.<sup>9</sup> Aligned collagen fibers supported the potential of metastasis of tumor cells.<sup>10</sup> Tumor cells were more invasive under the constraint of radial-patterned ECM.<sup>11</sup> Furthermore, fibrous collagen deposition and its crosslinking made the matrix denser.<sup>12</sup> The Ras-mitogen-activated protein kinase pathway was activated by higher matrix density to enhance the proliferation behavior of mammary epithelial cells.<sup>13</sup> The increase in the density of ECM also could inhibit the collective invasion of MCF-7 cells.<sup>14</sup> Meanwhile, denser ECM impeded migration of T cells and forced them change the direction of movement due to no production of proteases by T cells.<sup>15</sup> Proliferation of T cells was inhibited by high-density ECM fibers.<sup>16</sup> Thus, the regulatory effects of the morphology of ECM could result in more complex interactions between tumor cells and immune cells. The study of how alterations in the morphology of ECM fibers affected this complicated process could contribute to search effective methods to impede the progression of tumors. Therefore, the research on the tumor-immune interaction needs to be explored urgently.

The above-mentioned unaddressed issues further limited the effective application of immunotherapy. Low response rate of immune checkpoint inhibitors was still a big challenge in the current clinical treatment. The work about exploring more reliable predictive biomarkers, spatial characteristics of tumors, and so forth was under way.<sup>17</sup> Better understanding of morphological regulation of ECM in the interactions between tumor development and antitumor immune response could provide the potential target to enhance the effectiveness of immunotherapy.

A computation hybrid model is suitable for solving the complex task of such type. Several types of computational models have been developed to address various aspects of cancer development.<sup>18–21</sup> The effect of cancer stem cell on the resultant growth behavior of the multicellular system was investigated based on cellular automata model to assess the effectiveness of cancer treatment.<sup>22</sup> The cell-center model was utilized to investigate the effect of the urokinase-type plasminogen activator system on ECM proteolysis for exploring invasion of cancer cells.<sup>23</sup> The multiscale agent based model was used to investigate the spread of precancerous fields in the colonic epithelium.<sup>24</sup> The effect of microstructural and biomechanical features of ECM on tumor invasion was investigated based on continuum porous media model to predict the response of malignant masses to novel therapeutic agents affecting directly the tumor microenvironment.<sup>25</sup> Of particular interest is the cellular potts model (CPM), a computational model used to simulate the collective behavior of cells, capable of simulating cell-cell and cell-ECM interactions.<sup>26</sup> The effect of current therapeutic approaches on the early stages of bladder tumor evolution and invasiveness, like chemotherapy, surgical ablation of the tumor and immunotherapy, was explored.<sup>27</sup> The effect of oxygen and pro-angiogenic factors on avascular and vascular tumor growth was investigated to explore how tumor-induced angiogenesis affected the growth rate, size, and morphology of simulated tumors.<sup>28</sup> The effect of collagen I gels on glioma cell lines was investigated to reproduce the dispersed glioma invasion patterns.<sup>29</sup> The effect of cell-cell adhesion on invasiveness of the tumor was investigated to demonstrate that the morphology of the invading front was influenced by changes in parameters related with adhesiveness.<sup>30</sup> The CPM-based model could use an effective potential formalism to describe the attributes of the discrete individuals and the rules for their dynamics and interactions with fields, which was a multiscale hybrid framework that was able to deal with the typical multilevel organization of biological development.<sup>31</sup>

In this work, a CPM-based model was developed to explore how the density and the morphology of ECM regulated the tumor-immunity interaction. The density of the ECM fibers was determined from the number of fibers within the simulation domain. Three morphologies, including random, linear, and radial patterns, were created. Migration of T cells, tumor cells, and transition efficiency of cell states were analyzed in the model. By linking the spatio-temporal features with adjustment of morphological behaviors of ECM, the proposed model provided a framework to analyze the potential role of ECM fibers involved with the effectiveness of antitumor immune response. Thus, ECM fiber could be regarded as a significant target combined with the application of immunotherapy to further improve the therapeutic effect of tumors.

## 2 | METHOD

### 2.1 | The principle of cellular potts model

A two-dimensional multiscale model was built based on the principle of CPM. Each cell was represented by generalized cells with a collection of lattice pixels and the specific cell ID. Each cell was given specific attributes, including volume, surface, chemotaxis, adhesion, and so forth.<sup>32</sup> The core of the CPM was effective energy, which described the individual behavior of cells and their interactions with other cells. The effective energy of the developed tumor cell–ECM–immune cell system was calculated on the basis of the Hamilton function<sup>33</sup>:

$$H = \sum_{ij} J(\tau(\sigma_i), \tau(\sigma_j)) (1 - \delta(\sigma_i, \sigma_j)) + \lambda_V \sum_i (V(\sigma_i) - V_T(\sigma_i))^2 + \lambda_S \sum_i (S(\sigma_i) - S_T(\sigma_i))^2 + \lambda_c \sum_i (c(x_{\text{source}}) - c(x_t)), \quad (1)$$

where  $J$  denoted the adhesion energy per unit contact length,  $\tau$  characterized the type of cells,  $\sigma$  was the label of cells,  $\delta$  was the Kronecker delta function,  $\lambda$  was the constraint strength that corresponds to elastic modulus,  $V$  was the instantaneous volume of cells,  $V_T$  denoted the target volume of cells,  $S$  was the instantaneous surface of cells,  $S_T$  denoted the target surface of cells, and  $c(x_{\text{source}})$  and  $c(x_t)$  denoted chemical concentration at the pixel-copy-source and pixel-copy-destination pixel, respectively.

Cell dynamics in the developed model was implemented using a stochastic modified metropolis algorithm and changes in index-copy attempts was achieved with the probability given by the Boltzmann acceptance function<sup>34</sup>:

$$P = \begin{cases} \exp\left(-\frac{\Delta H}{T_m}\right), & \Delta H > 0 \\ 1, & \Delta H \leq 0 \end{cases}, \quad (2)$$

where  $\Delta H$  denoted change in the effective energy of each copy attempt, and  $T_m$  represented the amplitude of cell membrane fluctuations. The Monte Carlo step iteratively increased after the index-copy attempts were completed in all the grid cells within the simulation domain.

### 2.2 | The overview of the proposed model

T cells, tumor cells, and ECM fibers were taken account into the model. Varying-density ECM fibers with random, aligned, and radial patterns were localized in the simulation domain, respectively. The density of ECM fibers was determined by the number of fibers within simulation domain. Tumor cells initially grew at the center of simulation domain, hypothetically controlled by the concentration of glucose. Glucose was secreted by surrounding medium. Its diffusion and consumption were followed by Equation (3).<sup>32</sup>

$$\frac{\partial G(x)}{\partial t} = D_G \nabla^2 G(x) + \delta(\tau(\sigma(x)), \text{tumor}) U_G^{\text{tumor}}(x) + \delta(\tau(\sigma(x)), \text{medium}) U_G^{\text{medium}}(x), \quad (3)$$

where,  $D_G$  was the diffusion rate of glucose, and  $U_G^{\text{medium}}$  and  $U_G^{\text{tumor}}$  denoted the uptake of glucose by medium and tumor cells, respectively.

Each tumor cell switched between proliferating, resting, and necrotic states depending on the glucose concentration at its current location. Above the concentration threshold of glucose, changes in the volumes of tumor cells were regulated by the concentrations of glucose, expressed as Equation (4), to characterized the growth of tumor cells.<sup>32</sup>

$$\frac{dV_T(\text{tumor})}{dt} = \frac{G_{\max} G(x)}{G(x) + G_0}, \quad (4)$$

where  $G_0$  was the concentration at which growth rate was half its maximum.

Activated T cells were recruited to the edge of the simulation domain at every time step and then migrated stochastically. The number of recruited activated T cells was associated with living tumor cells,<sup>33</sup> indicated as Equation (5).

$$\frac{dN_{AT}}{dt} = \frac{\rho N_T}{s + N_T} N_{AT}, \quad (5)$$

where  $N_{AT}$  was the number of live Activated T cells,  $N_T$  denoted the number of live tumor cells,  $\rho$  was maximum rate of recruited T cells, and  $s$  was the constant related with recruiting T cells.

Random-moving T cells could sense the concentration of antigens released by necrotic cells and then move toward tumor cells against the antigen gradient. The distribution pattern of antigens could be obtained as follows:

$$\frac{\partial A}{\partial t} = D_A \nabla^2 A + kA + S_A, \quad (6)$$

where  $D_A$  was the diffusion constant of antigens,  $k$  was the decay constant of antigens, and  $S_A$  denoted secretion amount of antigens.

Some motile T cells contacted with tumor cells and were transformed into cytotoxic T cells and formed tumor-effector complexes to further attract more activated T cells come over. Cytotoxic T cells killed tumor cells with the probability  $k_1$ . Tumor cells that were not killed by cytotoxic T cells could be transformed into PD-L1 positive tumor cells. PDL1+ tumor cells could inhibit the capability of cytotoxic T cells with the probability  $k_2$ .

In between tumor cells and activated T cells were ECM fibers. Random, aligned, and radial fibers with the variable number were created, respectively, to block the interactions between tumor cells and T cells. During the process of growth, tumor cells could contact with fibers and then secrete matrix metalloproteinases (MMP) to dissolve fibers. The dynamic distribution of MMP was shown in Equation (7). Activated T cells passed through the interstitial space of ECM fibers. Variable antitumor immune response could be observed.

$$\frac{\partial \text{MMP}(x,t)}{\partial t} = D_{\text{MMP}} \nabla^2 \text{MMP}(x,t) - \delta_{\text{MMP}} \text{MMP}(x,t) + S_{\text{MMP}}, \quad (7)$$

where,  $D_{\text{MMP}}$  was the diffusion constant of MMP,  $\delta_{\text{MMP}}$  was the degradation rate of MMP and  $S_{\text{MMP}}$  denoted the secretion term of MMP.

In the proposed model, the dimension of the simulation domain in pixels was 500\*500. One pixel in our simulation was set to 4  $\mu\text{m}$ . Therefore, the size of the simulation domain represented 2000  $\mu\text{m}$ , the size of each tumor cell in this simulation was 16  $\mu\text{m}^2$ , and activated T cells and cytotoxic T cells had the dimension of 12  $\mu\text{m}^2$ . ECM lattice with varying number and direction were created. The detailed information about organization of ECM fibers was shown in Reference [14]. The model was implemented using CompuCell3D. The values of model parameters were listed in Table 1.

### 3 | RESULTS

The proposed model intended to analyze how the density and the morphology of ECM fibers regulated behaviors of tumor cells and T cells and their interactions. Figure 1 showed the schematic of the model with varying-density and morphology ECM fibers. ECM fibers played the role of natural physical barriers to mediate the time period of occurrence and the intensity of the tumor-immune interaction. The developed model was validated qualitatively and quantitatively, indicated in the supplementary material.

Random ECM fibers with variable densities (400/ $\text{mm}^2$ , 1000/ $\text{mm}^2$ ) were localized in the simulation domain, respectively. Tumor cells initially grew in the center of simulation domain. As the increase of the number of tumor cells, the necrotic region of the tumor appeared due to lack of glucose in the center, leading more and more tumor cells migrated outward for sufficient glucose. Variations in the densities of ECM fibers regulated the interactions between tumor cells and T cells, as shown in Figure 2(A,B). As the time went on, tumor cells were gradually split into several scattered tumor masses. Until 10,000 MCS, tumor cells surrounded by ECM fibers with the lower density were almost split into independent tumor masses. Surrounded by high-density of ECM fibers, four tumor lumps were formed until 15,000 MCS. It was probably due to lack of glucose in the center or be killed by cytotoxic T cells. After 3700 MCS, the

**TABLE 1** Values of parameters used in simulations

| Parameter                                                                                 | Value                                                           | Remark   |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------|
| Lattice size                                                                              | $2 \times 2 \text{ mm}^2$ ( $500 \times 500 \text{ pixels}^2$ ) | [14]     |
| Volume per voxel                                                                          | $64 \mu\text{m}^3$                                              | [32]     |
| Initial target volume ( $V_T$ )                                                           | 16                                                              | [35]     |
| Initial target surface ( $S_T$ )                                                          | 16                                                              | [35]     |
| Lambda volume ( $\lambda_V$ )                                                             | 15                                                              | [35]     |
| Lambda surface ( $\lambda_S$ )                                                            | 5                                                               | [35]     |
| Glucose diffusion constant ( $D_G$ )                                                      | $13,500 \text{ pixels}^2/\text{MCS}$                            | [32]     |
| Glucose decay constant ( $U_G^{\text{medium}}$ )                                          | $.45 \text{ MCS}^{-1}$                                          | [32]     |
| Uniform glucose secretion rate                                                            | $.5 \text{ fmol/voxel/MCS}$                                     | [35]     |
| QCancer glucose max. uptake rate ( $G_{Q\text{max}}$ )                                    | $1.69 \text{ fmol/voxel/MCS}$                                   | [32]     |
| QCancer Michaelis–Menten coefficient ( $G_0$ )                                            | $.00256 \text{ fmol/voxel}$                                     | [32]     |
| PCancer glucose max. uptake rate ( $G_{P\text{max}}$ )                                    | $2.25 \text{ fmol/voxel/MCS}$                                   | [32]     |
| PCancer Michaelis–Menten coefficient ( $G_0$ )                                            | $.00256 \text{ fmol/voxel}$                                     | [32]     |
| Recruite T cells max. rate ( $\rho$ )                                                     | $2 \text{ MCS}^{-1}$                                            | [36]     |
| The constant related with recruiting T cells ( $s$ )                                      | 20                                                              | [36]     |
| Antigens diffusion constant ( $D_A$ )                                                     | $2000 \text{ pixels}^2/\text{MCS}$                              | [36]     |
| Antigens decay constant ( $k$ )                                                           | $.00001 \text{ MCS}^{-1}$                                       | [36]     |
| Antigens secretion rate ( $S_A$ )                                                         | $10.2 \text{ MCS}^{-1}$                                         | [36,37]  |
| MMP diffusion constant ( $D_{\text{MMP}}$ )                                               | $.0027 \text{ pixels}^2/\text{MCS}$                             | [14,38]  |
| MMP decay constant ( $\delta_{\text{MMP}}$ )                                              | $.002 \text{ MCS}^{-1}$                                         | [14,39]  |
| MMP secretion rate ( $S_{\text{MMP}}$ )                                                   | $.01 \text{ MCS}^{-1}$                                          | [14]     |
| Probability of Cytotoxic T cells to kill tumor cells ( $k_1$ )                            | .8                                                              | [17]     |
| Probability of PDL1+ tumor cells to inhibit the capability of cytotoxic T cells ( $k_2$ ) | .5                                                              | [40]     |
| ECM fiber density                                                                         | 400 and $1000/\text{mm}^2$                                      | [5,41]   |
| ECM fiber orientation                                                                     | Random, radial, and aligned                                     | [4–6,42] |

Abbreviations: QCancer, quiescent cancer cells; PCancer, proliferative cancer cells.

consumption of glucose gradually became less in the condition of low-density ECM fibers. Dead cells could secrete antigens before they disappeared to further attract more activated T cells to enter the region of tumor cells. Low-density ECM fibers promoted obvious secretion of antigens at the edge of the tumor mass at the 3700th MCS. More and more antigens were produced in the subsequent process. Moreover, tumor cells had a higher probability of contacting with surrounding ECM fibers with the high density. Therefore, more ECM fibers could be dissolved by MMP secreted by tumor cells, which could dynamically regulate the antitumor immune response. The details of density-regulated antitumor immune response were discussed in the following parts.

Our study exhibited that more activated T cells translocated to the region of tumors for low-density ECM fibers, as shown in Figure 3(A). This result is consistent with previous experimental studies indicating that denser ECM impeded migration of T cells and forced them change the direction of movement due to no production of proteases by T cells.<sup>15</sup> When they contacted with tumor cells, activated T cells were transformed into cytotoxic T cells and then killed tumor cells to inhibit the further inside-out invasion of tumor cells. Strong inhibitory role of high-density ECM fibers made low conversion efficiency of activated T cells to cytotoxic T cells, as shown in Figure 3(B). Density-regulated T cells responses could then affect responses of tumor cells.

The maximum migration distance of tumor cells was defined as the maximum offset from the initial center of the tumor cluster per MCS. Figure 4(A) exhibited the maximum migration distance of tumor cells surrounded by varying-density ECM fibers. High-density ECM fibers prevented tumor cells from moving around more obviously than low-density ECM fibers did from 4700 to around 18,000 MCS, as directly exhibited by smaller dotted circles (green and red



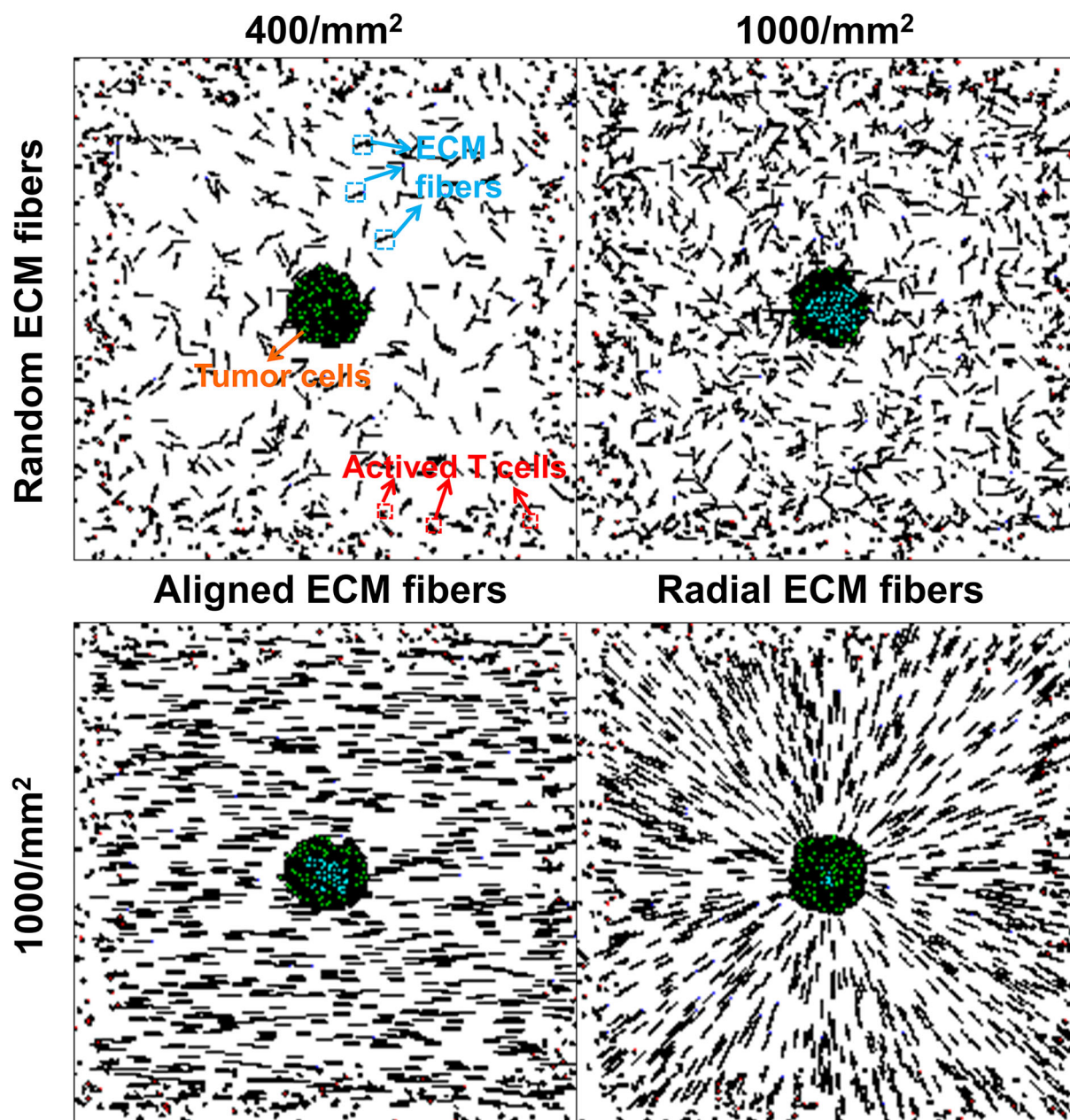
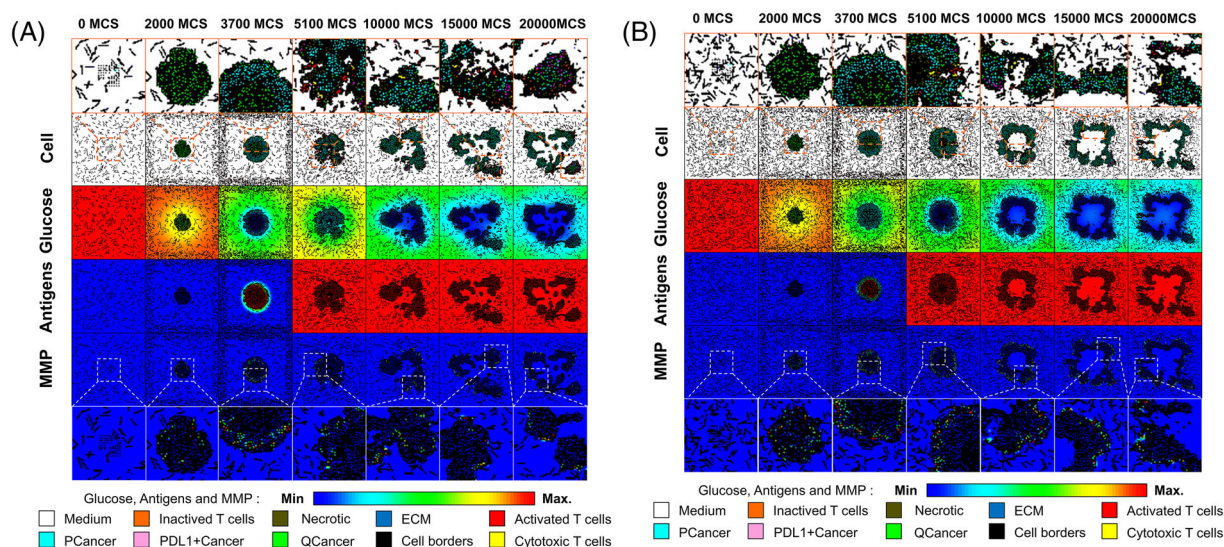
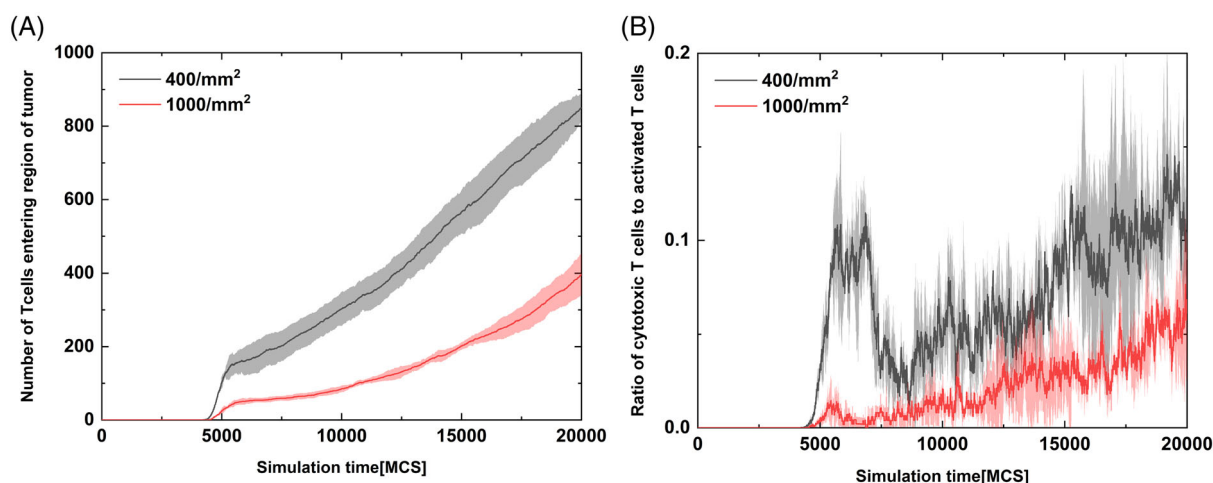


FIGURE 1 The schematic of the model with varying-density and morphology ECM fibers

for 10,000 and 15,000 MCS, respectively), which is consistent with the modeling result that increased ECM density for randomly aligned fibers confine the migration of tumor cells reported in Reference [14]. After 18,000 MCS, the maximum offset reached steady-state equilibrium (around 93  $\mu\text{m}$ ) for the ECM fibers with the density of 400/mm<sup>2</sup>. Smaller differences in maximum migration distance of T cells in both cases, which indicated that the blocking function of ECM fibers in tumor cells mainly occurred in the early stages. Changes in the number of all living tumor cells were indicated in Figure 4(B). At around 5100 MCS, there were significant downward trends, especially for low-density ECM fibers. Then, the growth rate of tumor cells slowed down over time. The number of living tumor cells for high-density ECM fibers was larger than that for low-density ECM fibers. Tumor cells that were not killed were transformed into PDL1+ tumor cells with probability  $k_2$ , as shown in Figure 4(C). PDL1+ tumor cells for low-density ECM fibers appeared earlier than those in the condition of high-density ECM fibers. The initiation of transformation from tumor cells that had not been killed to PDL1+ tumor cells was inhibited by high-density ECM fibers. Around 5000 MCS, there was an obvious peak of the number of PDL1+ tumor cells for ECM fibers with a density of 400/mm<sup>2</sup>. The decrease of total living tumor cells and occurrence of PDL1+ tumor cells inhibited the recruitment and translocation of activated T cells



**FIGURE 2** The dynamic changes in tumor cell–T cell interactions for random ECM fibers with a density of 400/mm<sup>2</sup> (A) and 1000/mm<sup>2</sup> (B)



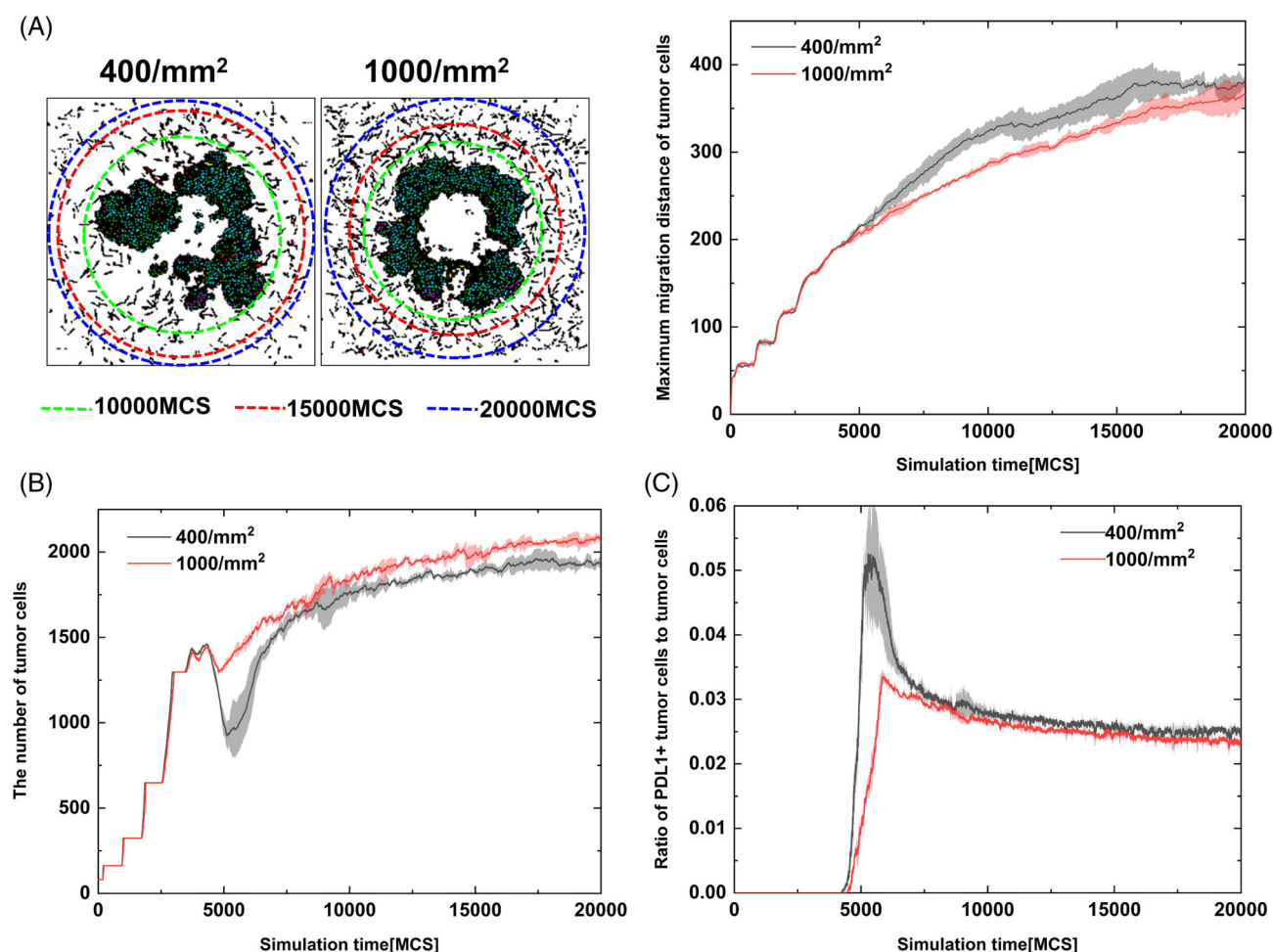
**FIGURE 3** Effect of ECM density on immune cells. (A) Changes in the number of T cells entering the region of tumor clustering. (B) Changes in ratio of cytotoxic T cells to activated T cells

corresponding to Figure 3(A). Therefore, tumor cells had chance of further increase due to limited recruited activated T cells and low conversion efficiency of cytotoxic T cells for low-density ECM fibers. After about 6000 MCS, no significant differences in conversion efficiency of PDL1+ tumor cells could be observed. The blocking function of ECM fibers could be weakened resulting from the facts that ECM fibers were gradually dissolved by MMP secreted by tumor cells and that loss of expression of PD-L1 on the surface of tumor cells with the specific probability, as demonstrated by previous studies.<sup>17</sup> Therefore, the density-regulation of ECM fibers on tumor cells and T cells occurred in the early stages. Given the equilibrium between tumor cells and T cells in the subsequent stages (steady amount of PDL1+ tumor cells and undifferentiated conversion efficiency of activated T cells to cytotoxic T cells), tumor cells reached the steady state.

Moreover, tumor development was accompanied by changes in the morphology of ECM fibers.<sup>4</sup> Random, aligned, and radial patterns were commonly observed near the region of tumors. In this work, these three architectures of ECM fibers with a density of 1000/mm<sup>2</sup> were created to explore the effect of ECM morphology on the interactions between tumor cells and antitumor immune response.

Variations in the morphology of ECM fibers regulated the interactions between tumor cells and T cells, as shown in Figure 5(A–C). For random ECM fibers, the tumor cluster spread outwards in a round shape and the cluster of activated





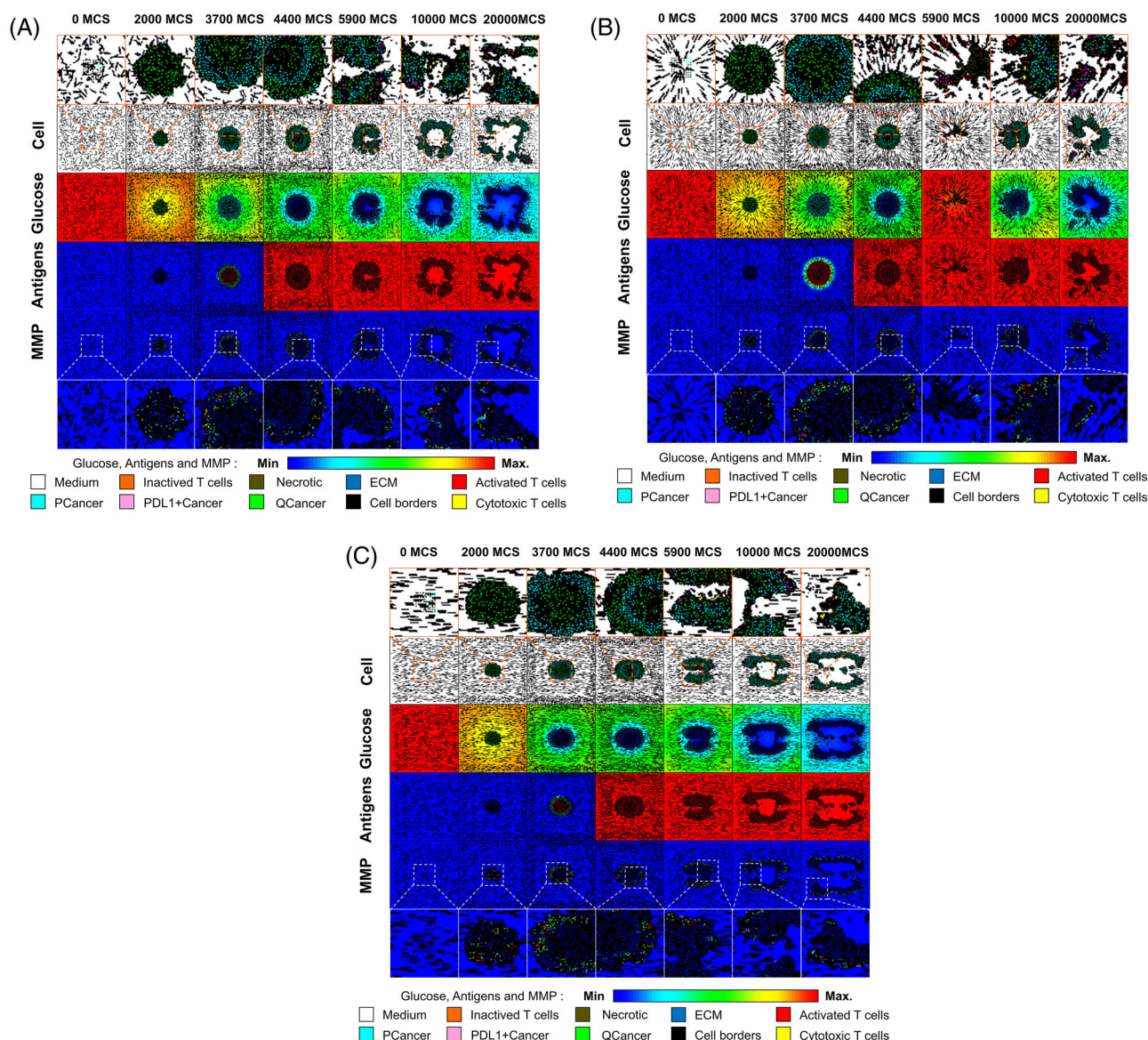
**FIGURE 4** Effect of ECM density on tumor cells. (A) Changes in the maximum migration distance of tumor clusters for varying-density random ECM fibers. (B) Changes in the number of living tumor cells. (C) Changes in ratio of PDL1+ tumor cells to all living tumor cells

T cells almost synchronously migrated from the edge of simulation domain to the center in a rectangular shape. For aligned ECM fibers, tumor cells, and activated T cells moved along the direction of fibers (lateral) and then interacted with each other. Surrounded by radial ECM fibers, the tumor gradually became a round mass and activated T cells migrated toward the region of tumors faster along the central area horizontally and longitudinally. As the time went on, tumor cells were gradually split into several scattered tumor masses. Until 5900 MCS, tumor cells surrounded by ECM fibers with the random and aligned patterns were gradually split into independent tumor masses, while the number of tumor cells significantly decreased in the case of radial ECM fibers. It was probably because of lack of glucose in the center or be killed by cytotoxic T cells. After 5900 MCS, the consumption of glucose became obviously less in the condition of radial-patterned ECM fibers. Radial-patterned ECM fibers promoted earlier secretion of antigens secreted by dead cells before they disappeared at the edge of the tumor mass than those with random and aligned patterns. More and more antigens were generated produced in the subsequent process. The details of morphology-regulated antitumor immune response were discussed in the following parts.

More activated T cells were translocated to the region of tumors for radial ECM fibers compared with random and aligned ECM fibers, as shown in Figure 6(A). High conversion efficiency of activated T cells to cytotoxic T cells in the case of radial ECM fibers could be observed before 17,500 MCS, as shown in Figure 6(B). No distinct differences in the number of translocated T cells and conversion efficiency of cytotoxic T cells could be exhibited for random and aligned ECM fibers. Therefore, the response of T cells could be morphology-specific.

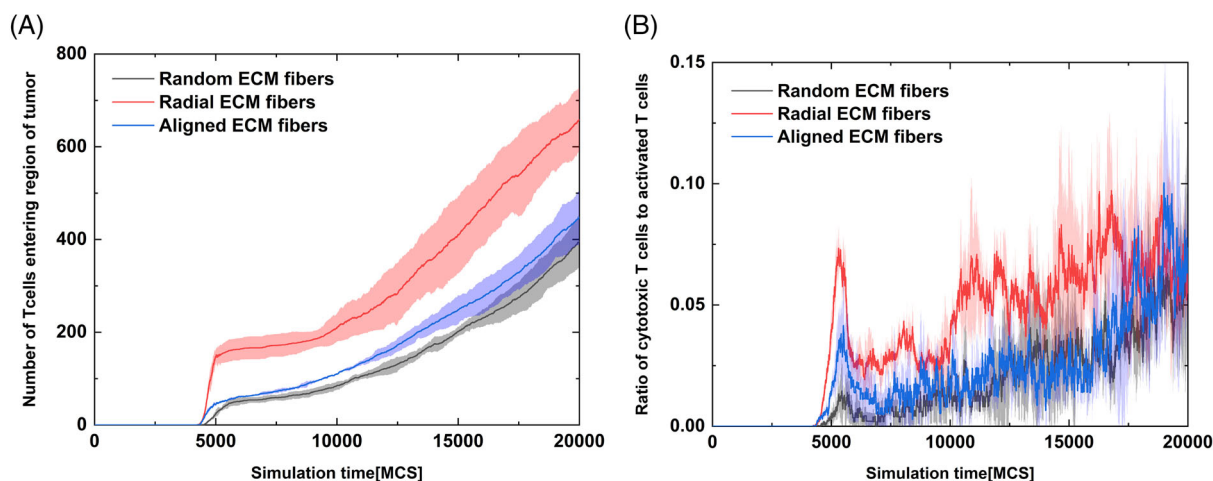
Figure 7(A) exhibited the maximum migration distance of tumor cells surrounded by varying-morphology ECM fibers. Before around 4700 MCS, changes in migration of tumor cells had no significant difference among these three ECM fibers. The blocking function of radial-patterned ECM fibers occurred weaker than the other two fibers



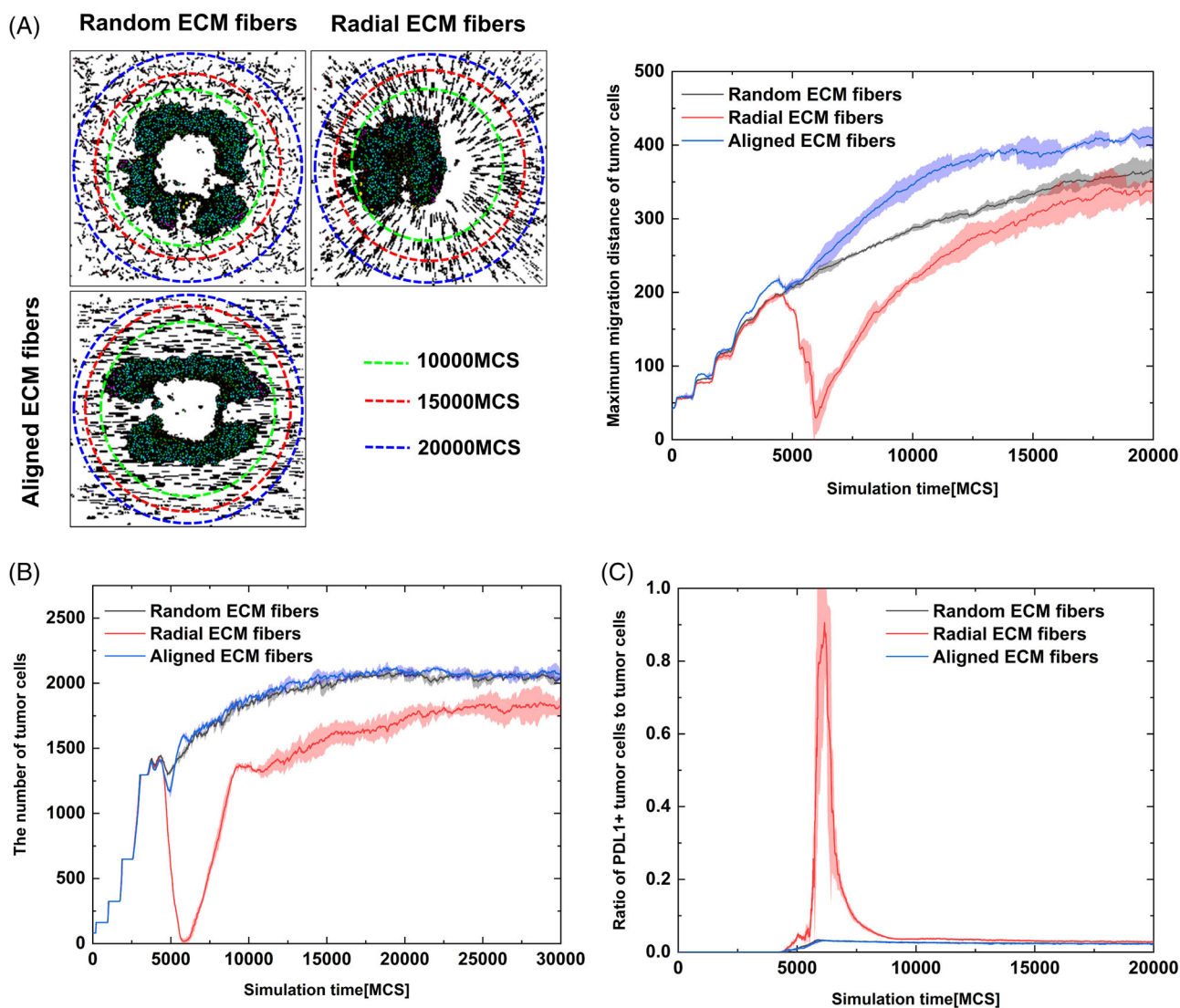


**FIGURE 5** The dynamic changes in tumor cell-T cell interactions for random ECM fibers (A), radial ECM fibers (B), and aligned ECM fibers (C)

to obviously inhibit the migration distance of tumor cells around 5000 MCS probably due to the killing capability of cytotoxic T cells, which was consistent with the fact that more cytotoxic T cells appeared near the region of tumors at that time, as shown in Figure 6(A). Aligned ECM fibers allowed longer migration distance of tumor cells than random ECM fibers did, corresponding to the higher rate of increase in migration distance from 6000 to 12,500 MCS. The migration distance of tumor cells gradually flattened out for aligned ECM fibers, while the migration distance of tumor cells for random ECM fibers gradually increased until 20,000 MCS. Thus, the blocking strength of random ECM fibers to tumor cells was stronger than that of aligned ECM fibers from 5000 to 10,000 MCS, corresponding to smaller migration distance and lower transform efficiency of activated T cells into cytotoxic T cells (Figure 6B). Around 5000 MCS, there was no difference in the number of tumor cells among three ECM fibers. As the time went on, the number of tumor cells surrounded by radial-patterned ECM fibers significantly decreased around 6000 MCS, as shown in Figure 7(B). However, the ratio of PDL1+ tumor cells to tumor cells in the case of radial morphology was larger than that in the other two cases due to higher probability of contact with T cells (Figure 7C). The occurrence of more PDL1+ tumor cells surrounded by radial ECM fibers inhibited the killing capability of cytotoxic T cells and resulted in the increase in the number of tumor cells. However, due to significant decrease of tumor cells for radial ECM fibers around 5000 MCS, the increase of the living tumor cells from



**FIGURE 6** Effect of ECM morphology on immune cells. (A) Changes in the number of T cells entering the region of tumor clustering. (B) Changes in ratio of cytotoxic T cells to activated T cells



**FIGURE 7** Effect of ECM morphology on tumor cells. (A) The maximum migration distance of tumor cells for ECM fibers with different patterns. (B) Changes in the number of tumor cells. (C) Changes in ratio of PDL1+ tumor cells to tumor cells



very low baseline was limited, so the number of living tumor cells for radial ECM fiber was lower than those in the other two cases.

## 4 | DISCUSSION AND CONCLUSIONS

Effective interactions between tumor cells and antitumor immune response affected the tumor development, which could be regulated by ECM fibers. In this work, morphological behaviors of ECM fibers were studied based on the computational modeling to explore the roles of them in interactions between tumor cells and T cells. The density and the morphology of ECM fibers affected the early stage of the above-mentioned interactions. For T cells, high-density ECM fibers prevented activated T cells from entering the region of tumor cells and decreased the conversion efficiency of activated T cells to cytotoxic T cells. For tumor cells, high-density ECM fibers prevented tumor cells from migrating outwards and decreased the conversion efficiency to PDL1+ tumor cells. Meanwhile, the number of tumor cells surrounded by high-density ECM fibers was larger than that for low-density ECM fibers. As for the effect of the morphology of ECM fibers, our study exhibited that more activated T cells translocated through radial ECM fibers and higher conversion efficiency of cytotoxic T cells mediated by radial ECM fibers. Meanwhile, the lower number of tumor cells and higher conversion efficiency of PDL1+ tumor cells could be observed under the condition of radial ECM fibers. Smaller migration distance of tumor cells also indicated weaker inhibitory effect of radial ECM fibers on interactions between tumor cells and immune cells. Therefore, in our work, based on the computational modeling, the ECM-regulated complex dynamic interactions between tumor cells and immune cells were elucidated more clearly. Our study demonstrated that morphological behaviors of ECM fibers were involved with the interaction between a tumor and the immune system, providing a potential foundation for further improving the efficiency of immune checkpoint inhibitor, such as PD-1/PD-L1, and so forth.

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